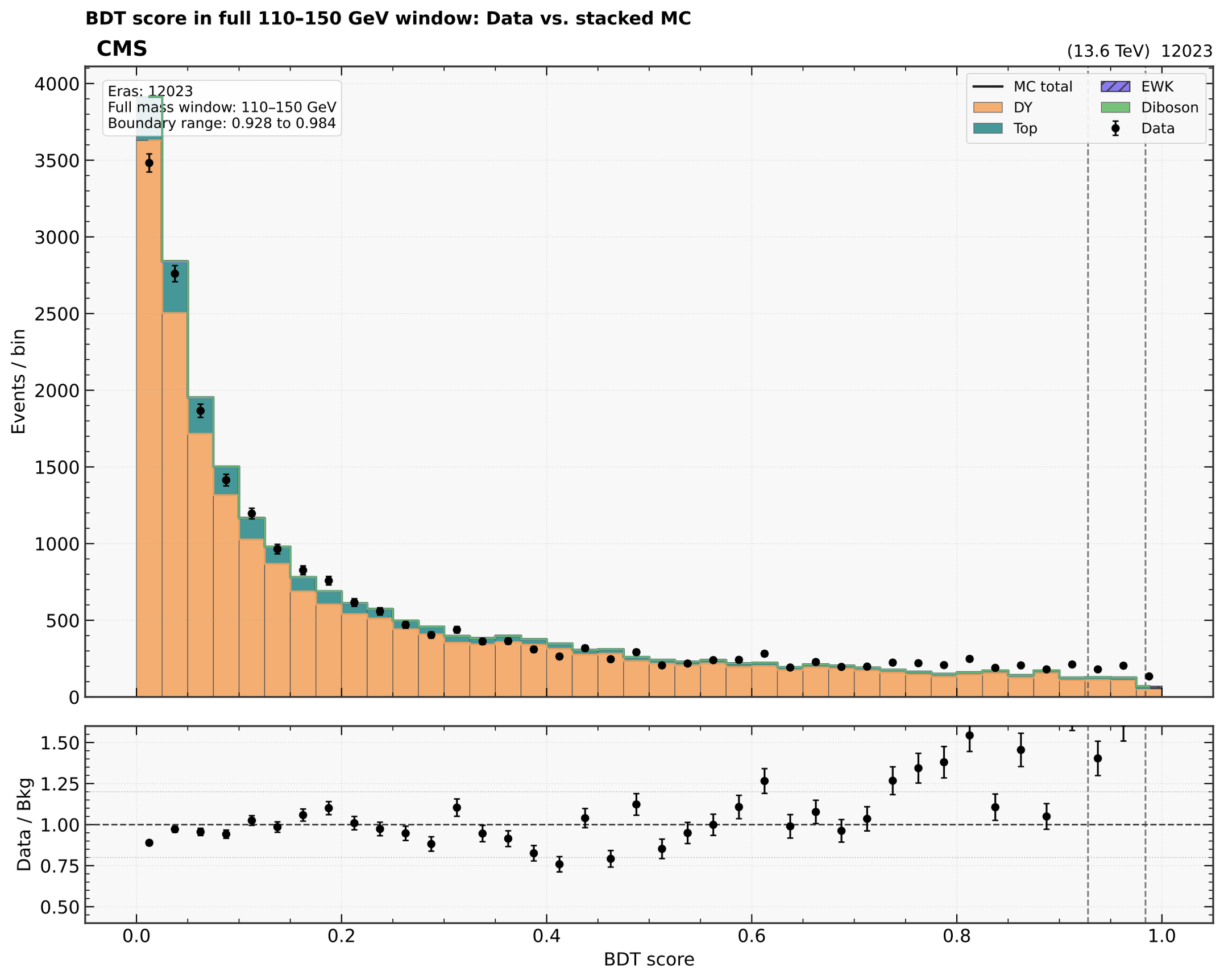
Run-3 H → μμ VBF BDT Category Optimisation

Eras : 12023
Category edges : 0.0000, 0.9279, 0.9636, 0.9840, 1.0000
MC scale factor : 1.0000 (Data/MC sideband normalisation)
Blinding window : 115–135 GeV
Sidebands : (110,115) ∪ (135,150) GeV
Outputs dir : /eos/home-m/memmad/HiggstoDimuon_VBF/src/drivers/9_TMVA/Outputs
Extra selection : –

CONTENTS:

p.2 BDT score distribution: Data vs. MC
p.3 BDT bin yields
p.4 Significance vs. number of categories
p.5 Boundary scan step-by-step (AN Fig. 32)
p.6 Atanh BDT + S/√B per category
p.7 BDT output vs. mass hypothesis (AN Fig. 26R)
p.8 Signal FWHM vs. BDT score
p.9 ROC curve with AUC
p.10 Blinded dimuon mass spectrum
p.11 Mass sculpting check (AN Fig. 25)
p.12 Per-category mass distributions and background fits
p.13 Background model comparison (χ²/ndf)
p.14 VBF signal DCB fits per category
p.15 DCB parameter table
p.16 σ_CB vs. category
p.17 Signal acceptance × efficiency
p.18 Per-category yield and sensitivity summary
p.19 Signal mass shapes mH=120/125/130 (AN Fig. 40)
p.20 Category summary table
p.21 Category process composition

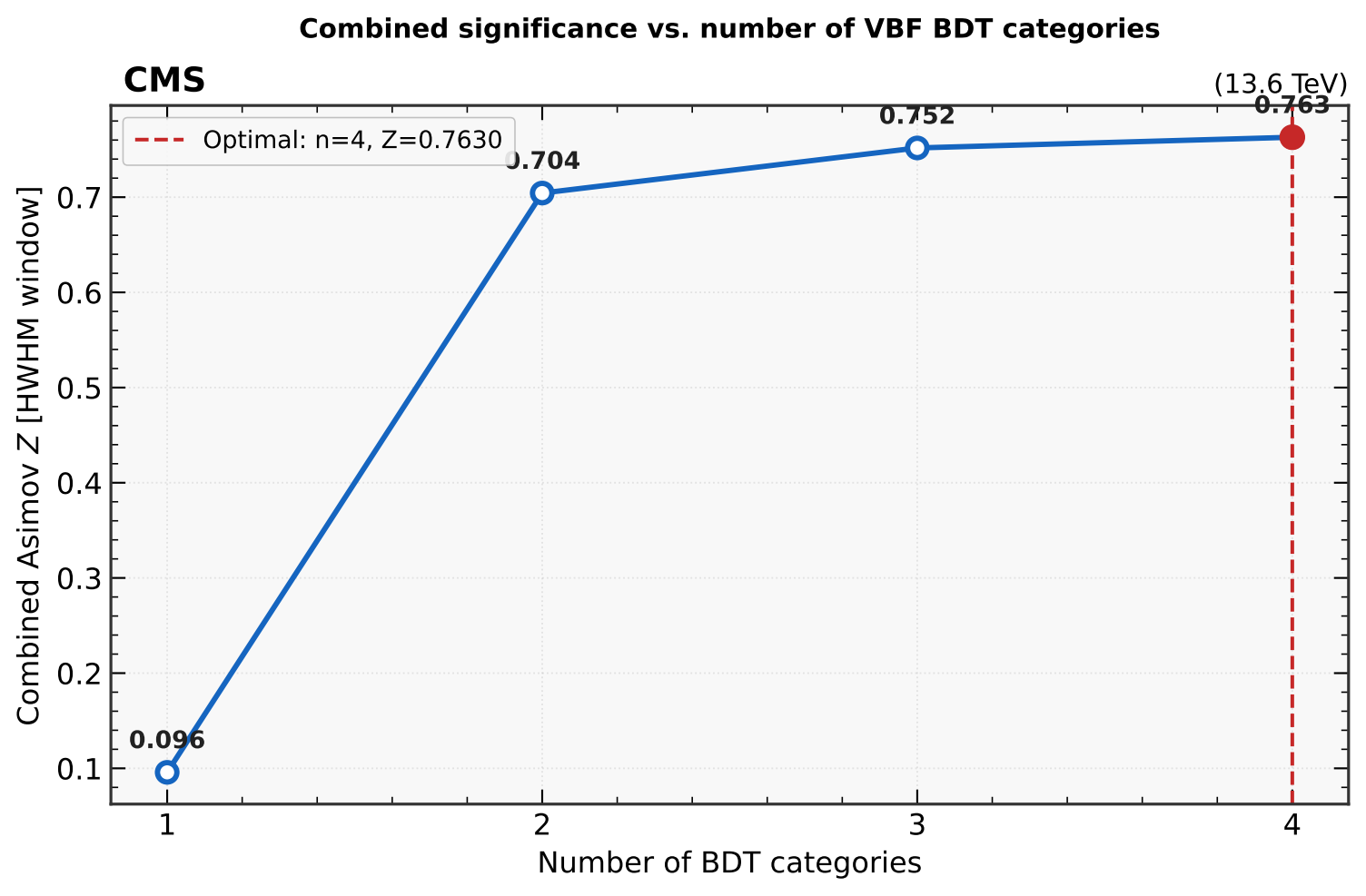

BDT score in full 110–150 GeV window: Data vs. stacked MC
CMS
(13.6 TeV) 12023
Eras: 12023
Full mass window: 110–150 GeV
Boundary range: 0.928 to 0.984
MC total
DY
Top
EWK
Diboson
Data
Events / bin
Data / Bkg
BDT score

BDT bin summary: signal, background, and sensitivity (page 1 / 2)

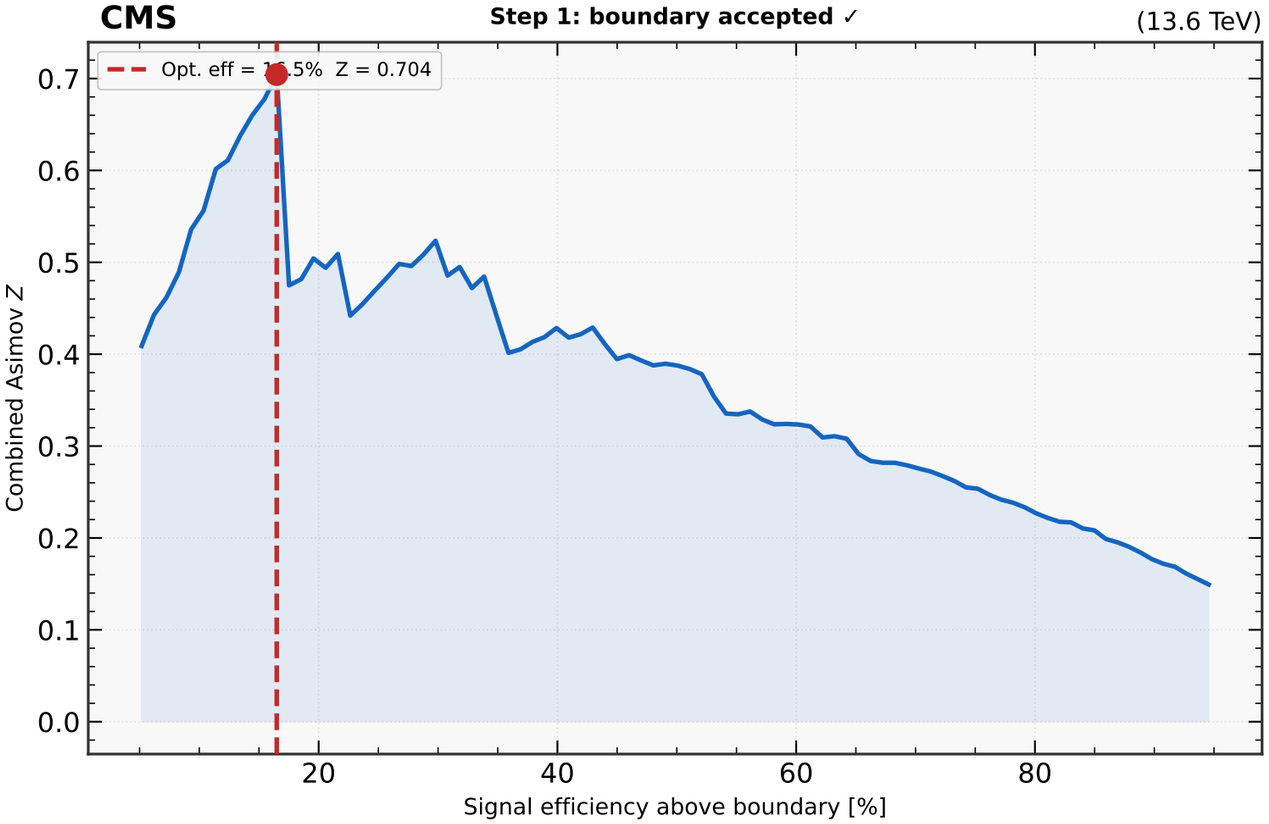
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.000, 0.025)	0.033	3915.80 ± 48.63	3482	0.00
[0.025, 0.050)	0.063	2840.38 ± 41.45	2760	0.00
[0.050, 0.075)	0.072	1952.95 ± 34.61	1866	0.00
[0.075, 0.100)	0.071	1501.68 ± 30.31	1414	0.00
[0.100, 0.125)	0.075	1166.82 ± 26.87	1196	0.01
[0.125, 0.150)	0.075	978.75 ± 24.82	964	0.01
[0.150, 0.175)	0.073	780.63 ± 22.00	826	0.01
[0.175, 0.200)	0.074	688.85 ± 20.48	758	0.01
[0.200, 0.225)	0.074	610.54 ± 19.75	616	0.01
[0.225, 0.250)	0.077	573.17 ± 19.44	558	0.01
[0.250, 0.275)	0.073	496.36 ± 18.09	470	0.01
[0.275, 0.300)	0.077	458.04 ± 16.96	404	0.02
[0.300, 0.325)	0.077	396.88 ± 15.84	438	0.02
[0.325, 0.350)	0.079	382.84 ± 15.53	362	0.02
[0.350, 0.375)	0.082	398.02 ± 16.02	364	0.02
[0.375, 0.400)	0.085	375.47 ± 15.54	310	0.02
[0.400, 0.425)	0.088	348.01 ± 14.95	264	0.03
[0.425, 0.450)	0.094	305.92 ± 14.19	318	0.03
[0.450, 0.475)	0.102	310.71 ± 14.30	246	0.03
[0.475, 0.500)	0.100	260.04 ± 12.90	292	0.04

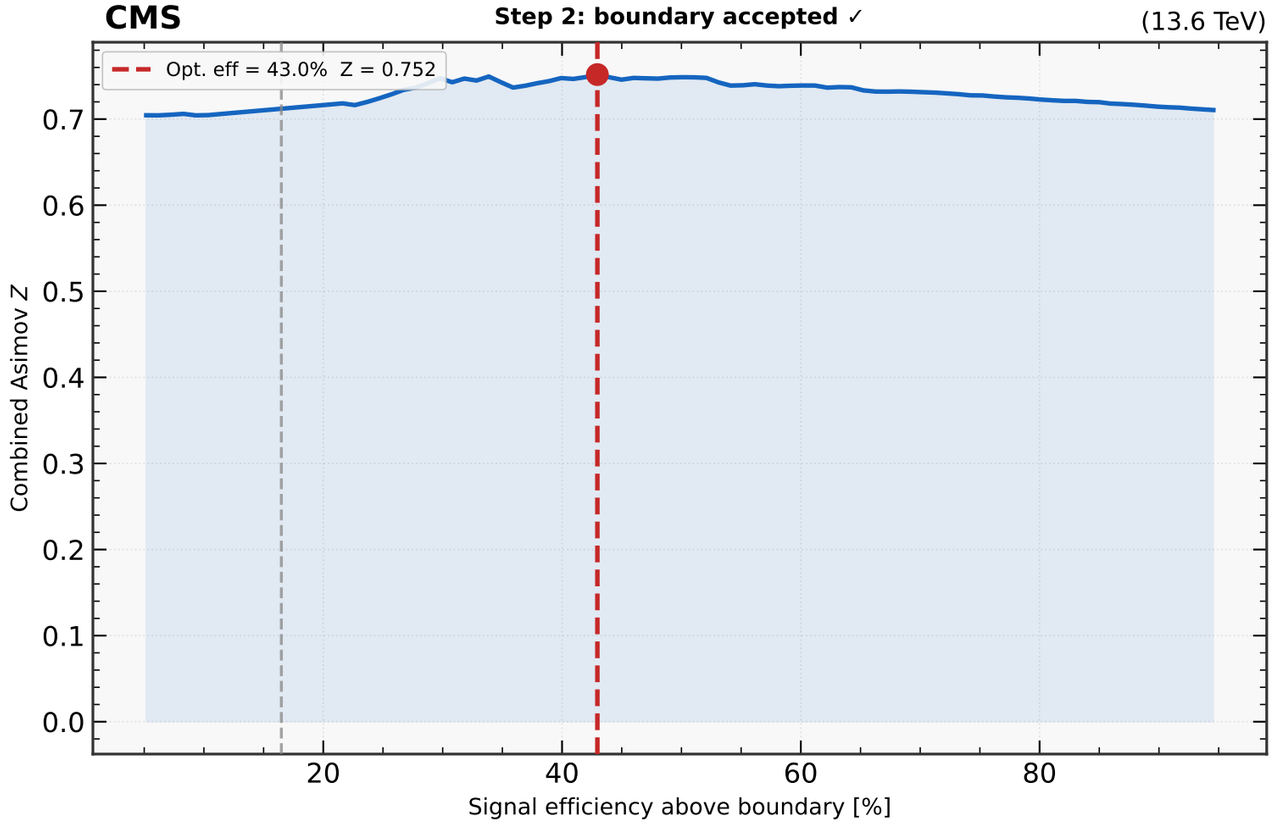
BDT bin summary: signal, background, and sensitivity (page 2 / 2)

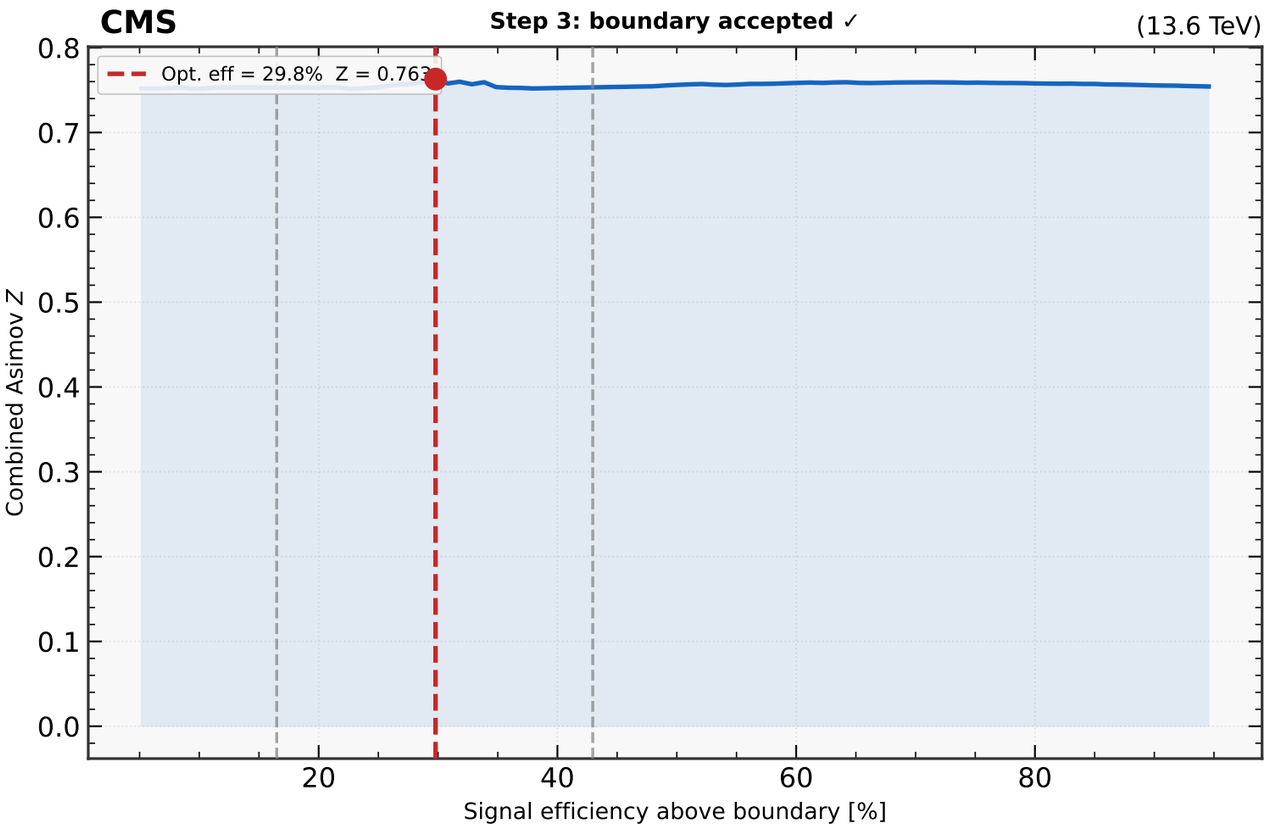
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.500, 0.525)	0.108	241.69 ± 12.43	206	0.04
[0.525, 0.550)	0.111	229.73 ± 12.11	218	0.05
[0.550, 0.575)	0.119	240.29 ± 12.41	240	0.05
[0.575, 0.600)	0.126	218.58 ± 11.67	242	0.06
[0.600, 0.625)	0.136	222.85 ± 12.20	282	0.06
[0.625, 0.650)	0.144	194.04 ± 11.09	192	0.07
[0.650, 0.675)	0.155	211.67 ± 11.66	228	0.07
[0.675, 0.700)	0.172	203.70 ± 11.31	196	0.08
[0.700, 0.725)	0.184	191.25 ± 10.89	198	0.10
[0.725, 0.750)	0.201	176.75 ± 10.72	224	0.11
[0.750, 0.775)	0.224	163.73 ± 10.06	220	0.14
[0.775, 0.800)	0.258	150.73 ± 9.49	208	0.17
[0.800, 0.825)	0.288	160.68 ± 10.24	248	0.18
[0.825, 0.850)	0.336	171.80 ± 10.46	190	0.20
[0.850, 0.875)	0.385	141.60 ± 9.55	206	0.27
[0.875, 0.900)	0.468	171.48 ± 10.47	180	0.27
[0.900, 0.925)	0.594	125.52 ± 8.64	212	0.47
[0.925, 0.950)	0.794	128.27 ± 8.98	180	0.62
[0.950, 0.975)	1.229	125.73 ± 8.73	204	0.97
[0.975, 1.000)	2.299	69.11 ± 6.09	134	3.22


Combined significance vs. number of VBF BDT categories
CMS
(13.6 TeV)
Optimal: n=4, Z=0.7630
0.096
0.704
0.752
0.763
Combined Asimov Z [HWHM window]
0.1
0.2
0.3
0.4
0.5
0.6
0.7
1
2
3
4
Number of BDT categories

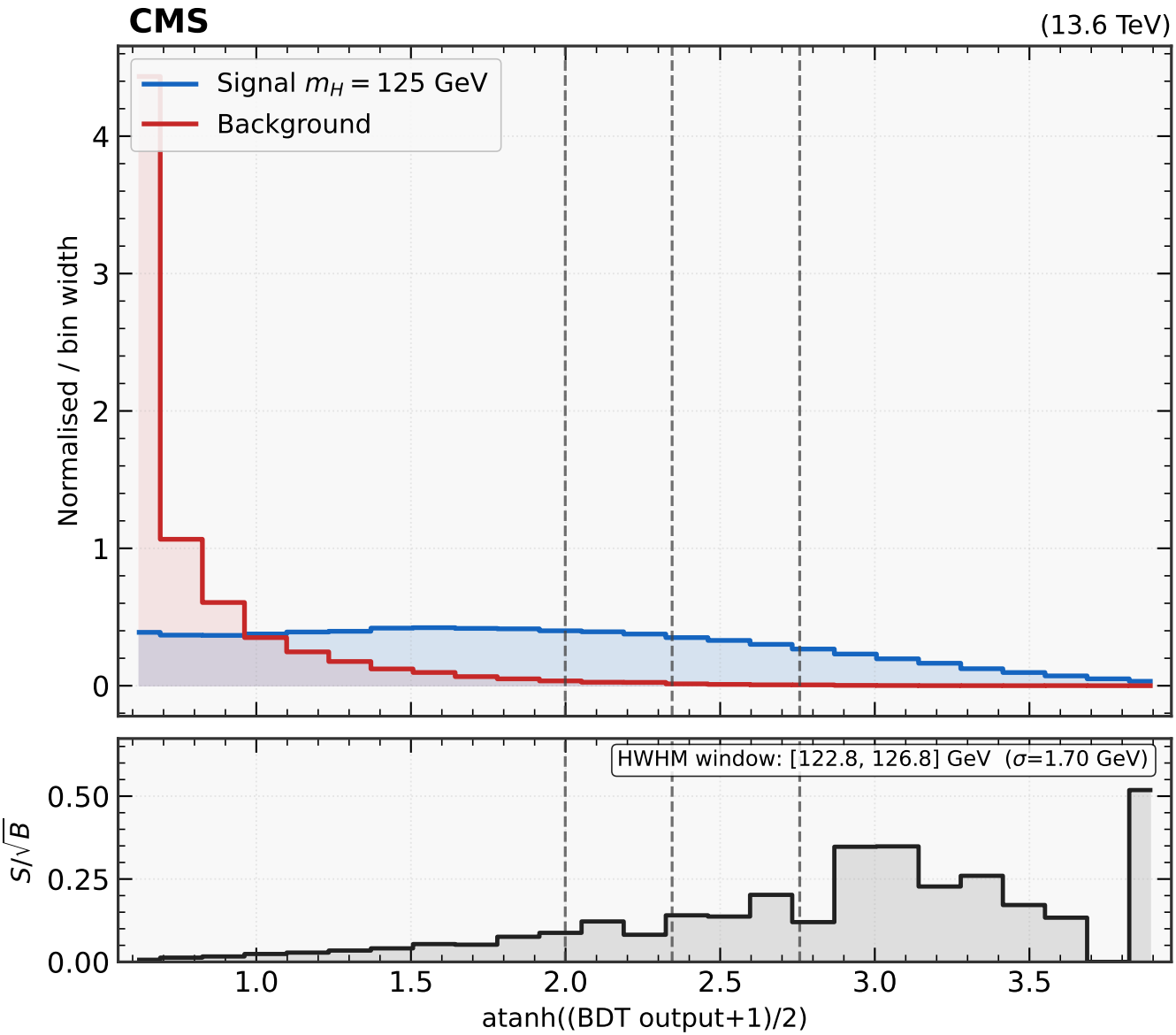
Iterative category boundary optimisation (AN Fig. 32 analog)

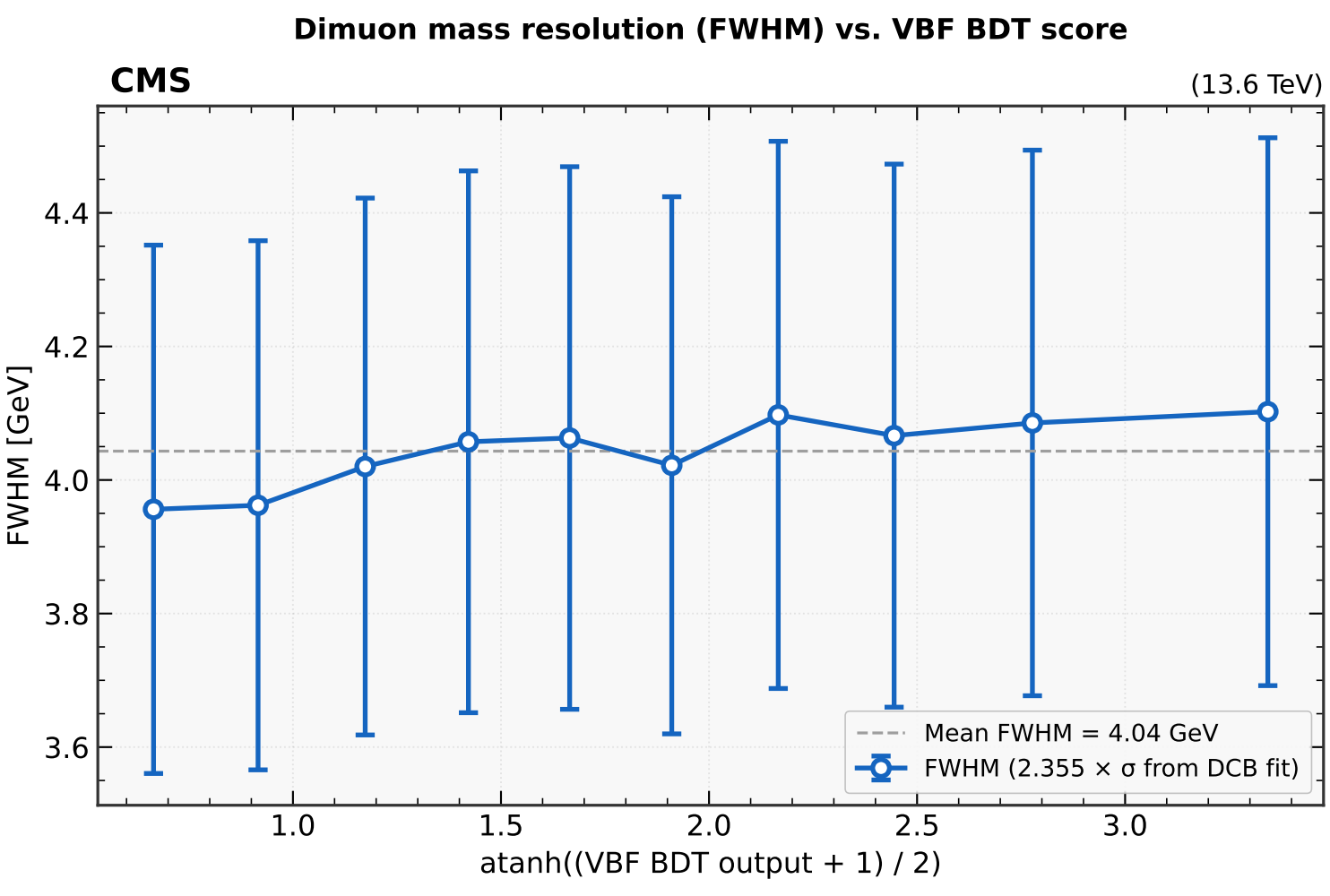


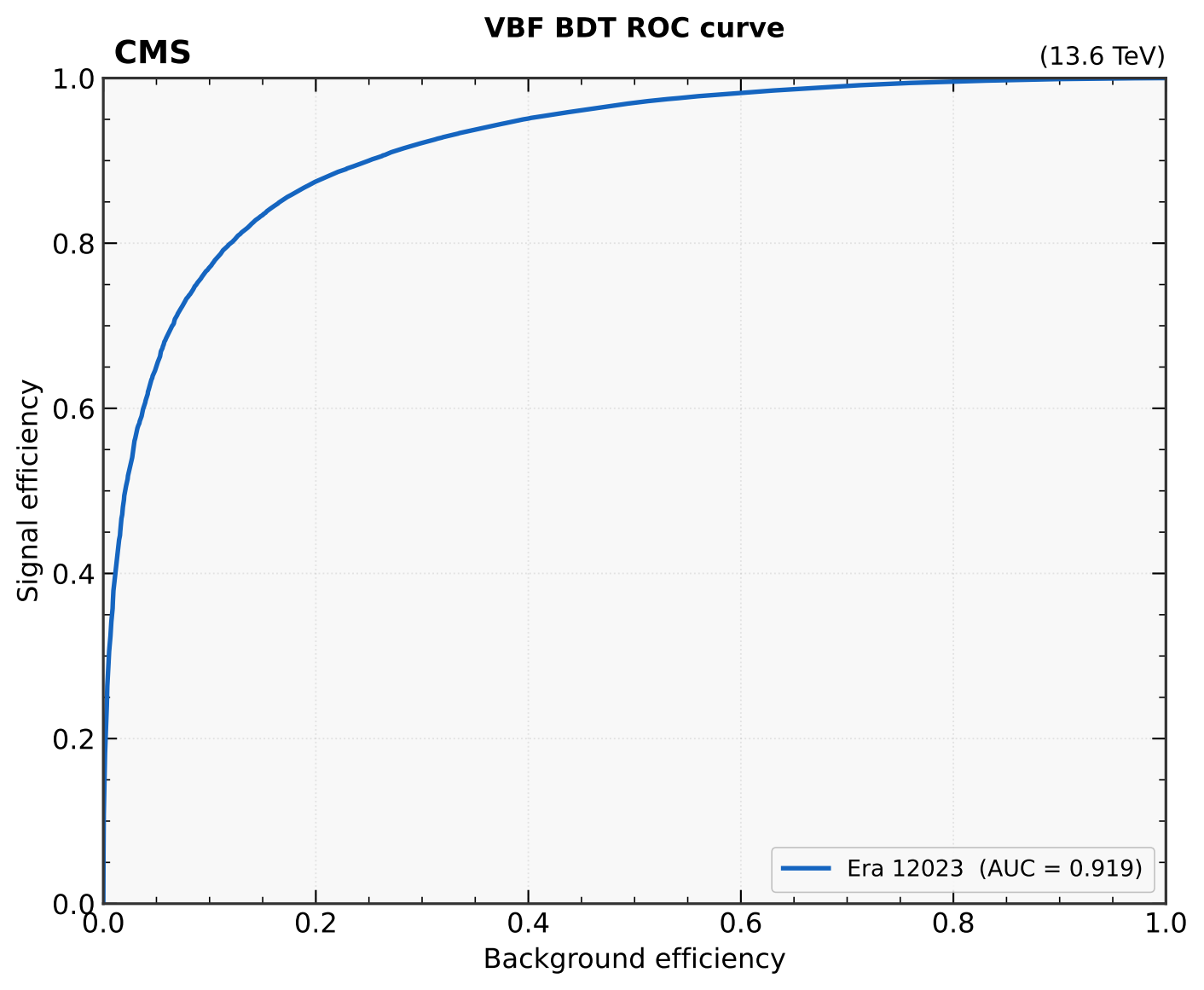


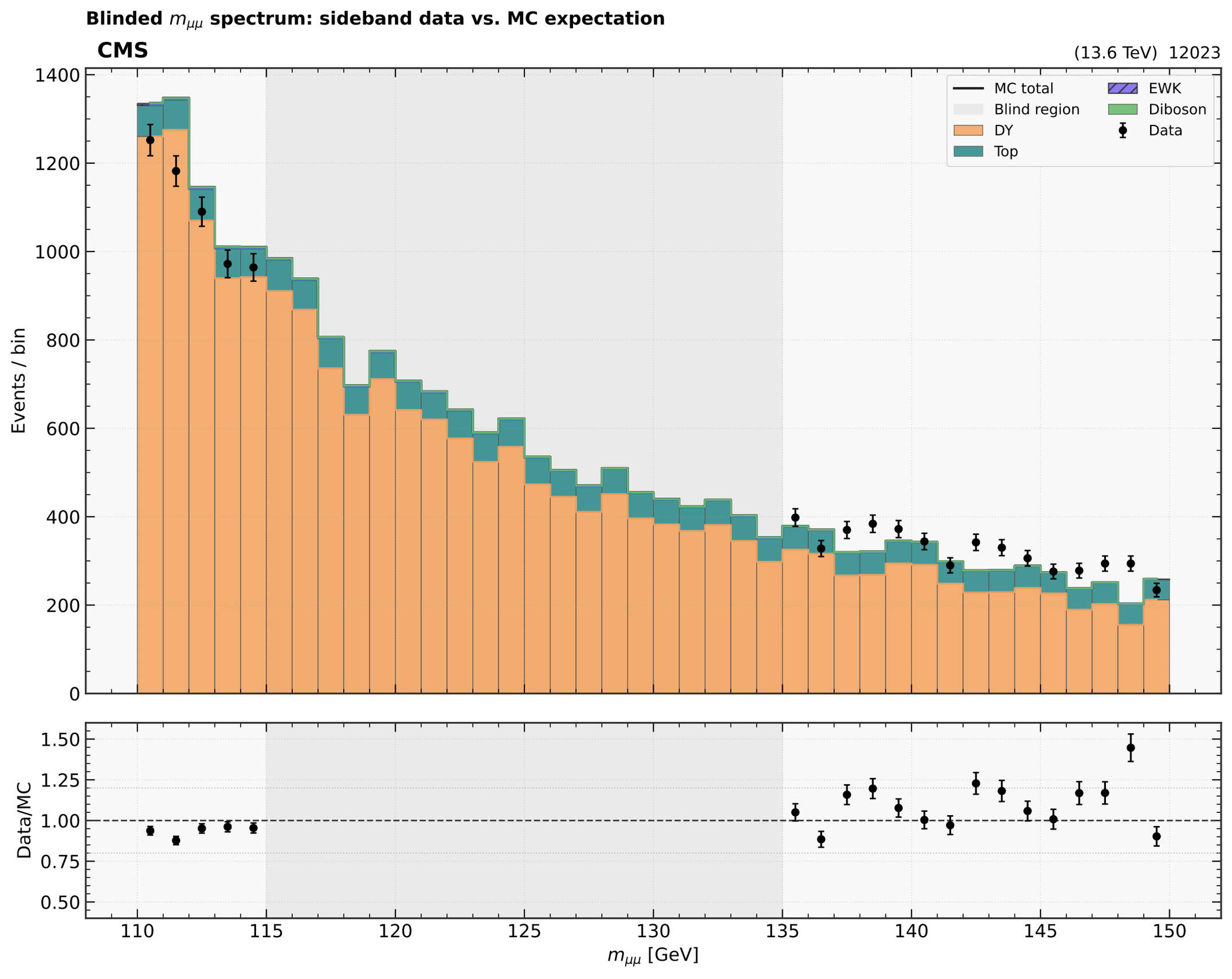


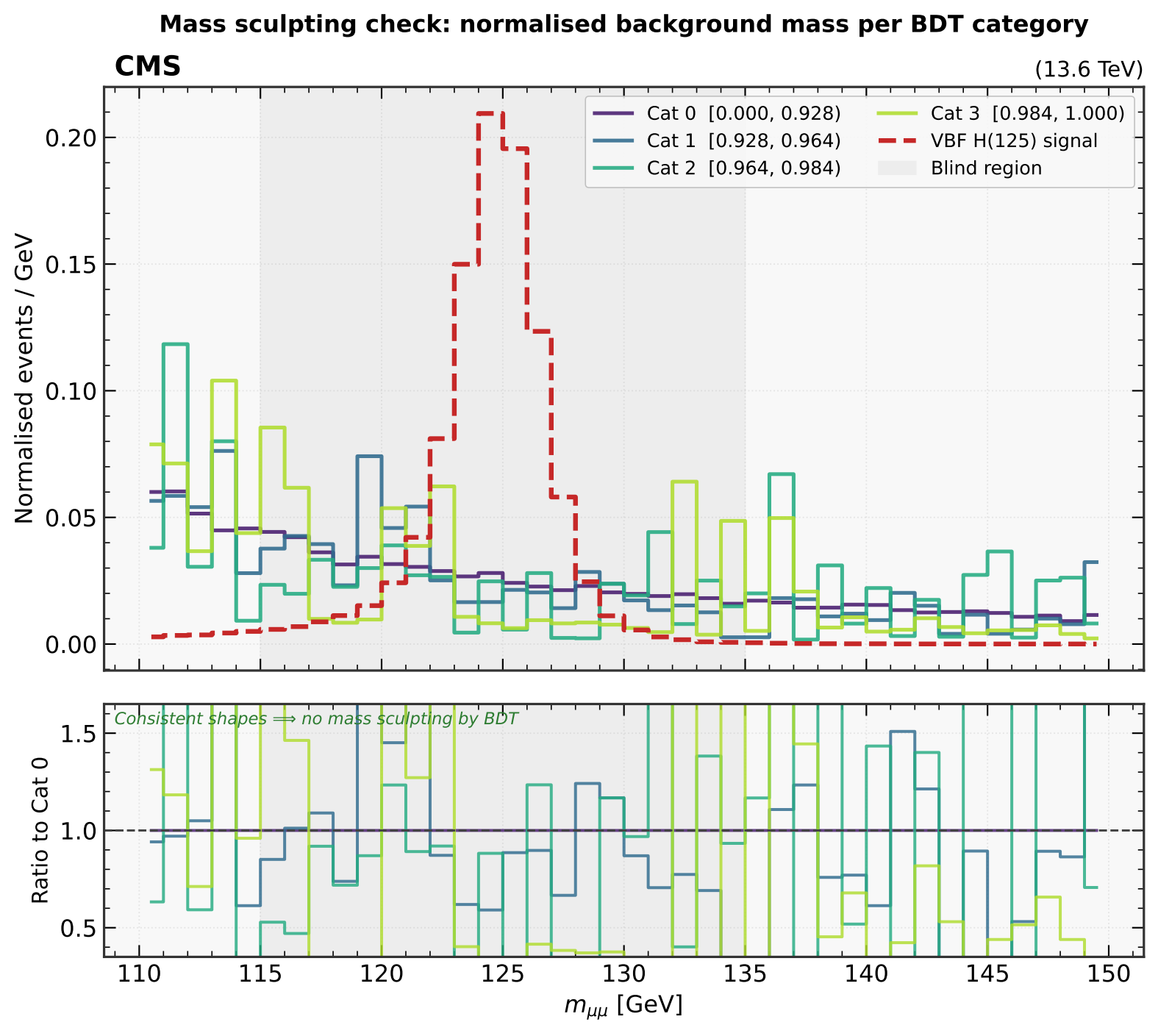
VBF BDT score and per-bin $S/\sqrt{B}$ in signal HWHM window




Dimuon mass resolution (FWHM) vs. VBF BDT score
CMS
(13.6 TeV)
FWHM [GeV]
4.4
4.2
4.0
3.8
3.6
1.0
1.5
2.0
2.5
3.0
atanh((VBF BDT output + 1) / 2)
Mean FWHM = 4.04 GeV
FWHM (2.355 × σ from DCB fit)

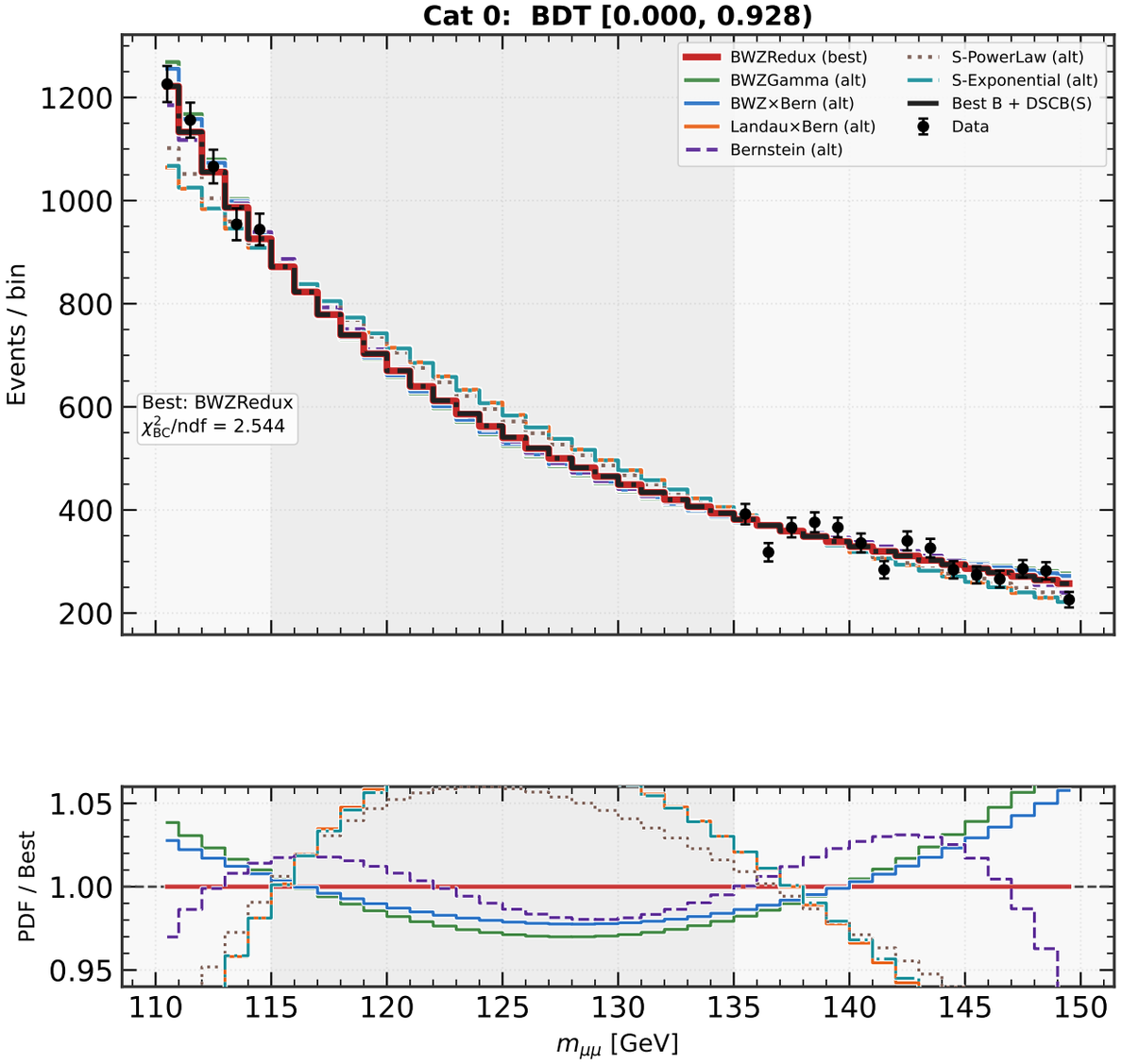

VBF BDT ROC curve
CMS
(13.6 TeV)
Signal efficiency
Background efficiency
0.0
0.2
0.4
0.6
0.8
1.0
Era 12023 (AUC = 0.919)


Blinded $m_{\mu\mu}$ spectrum: sideband data vs. MC expectation
CMS
(13.6 TeV) 12023
MC total
Blind region
DY
Top
EWK
Diboson
Data
Events / bin
Data/MC
$m_{\mu\mu}$ [GeV]

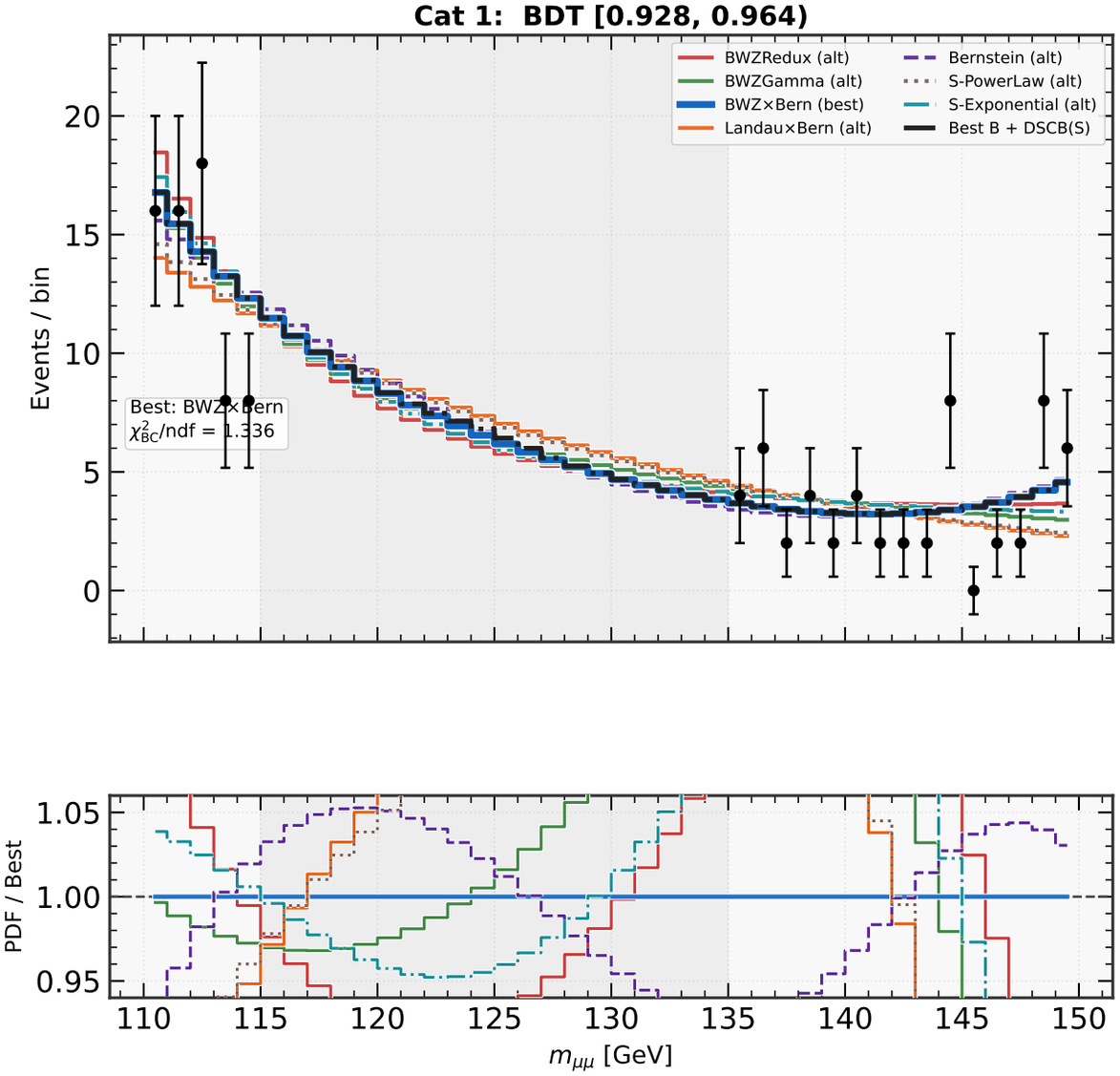

Mass sculpting check: normalised background mass per BDT category
CMS
(13.6 TeV)
Cat 0 [0.000, 0.928)
Cat 1 [0.928, 0.964)
Cat 2 [0.964, 0.984)
Cat 3 [0.984, 1.000)
VBF H(125) signal
Blind region
Normalised events / GeV
0.20
0.15
0.10
0.05
0.00
Consistent shapes ⟹ no mass sculpting by BDT
Ratio to Cat 0
1.5
1.0
0.5
110
115
120
125
130
135
140
145
150
$m_{\mu\mu}$ [GeV]

Per-category dimuon mass: background model fits

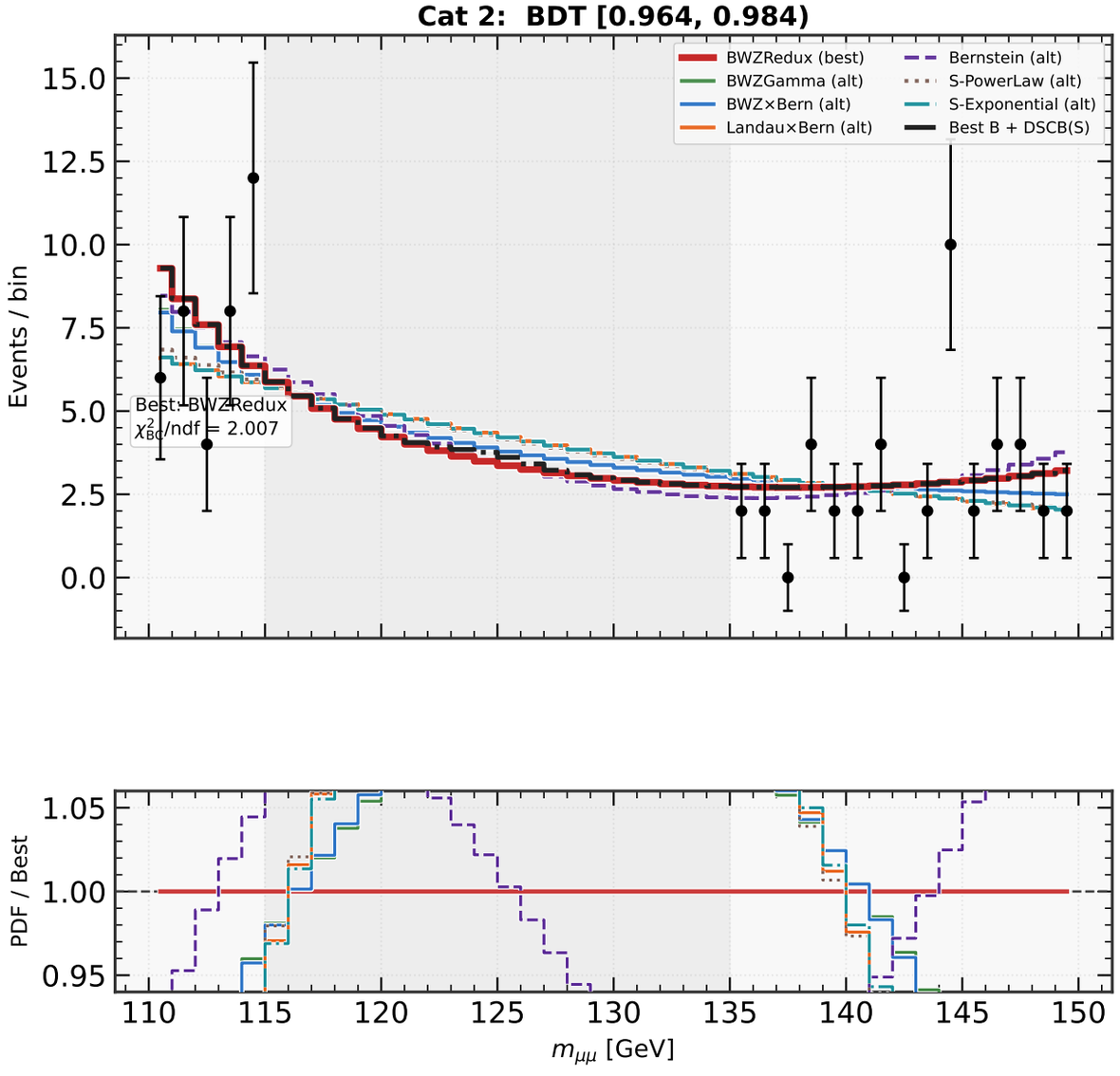
Cat 0: BDT [0.000, 0.928)



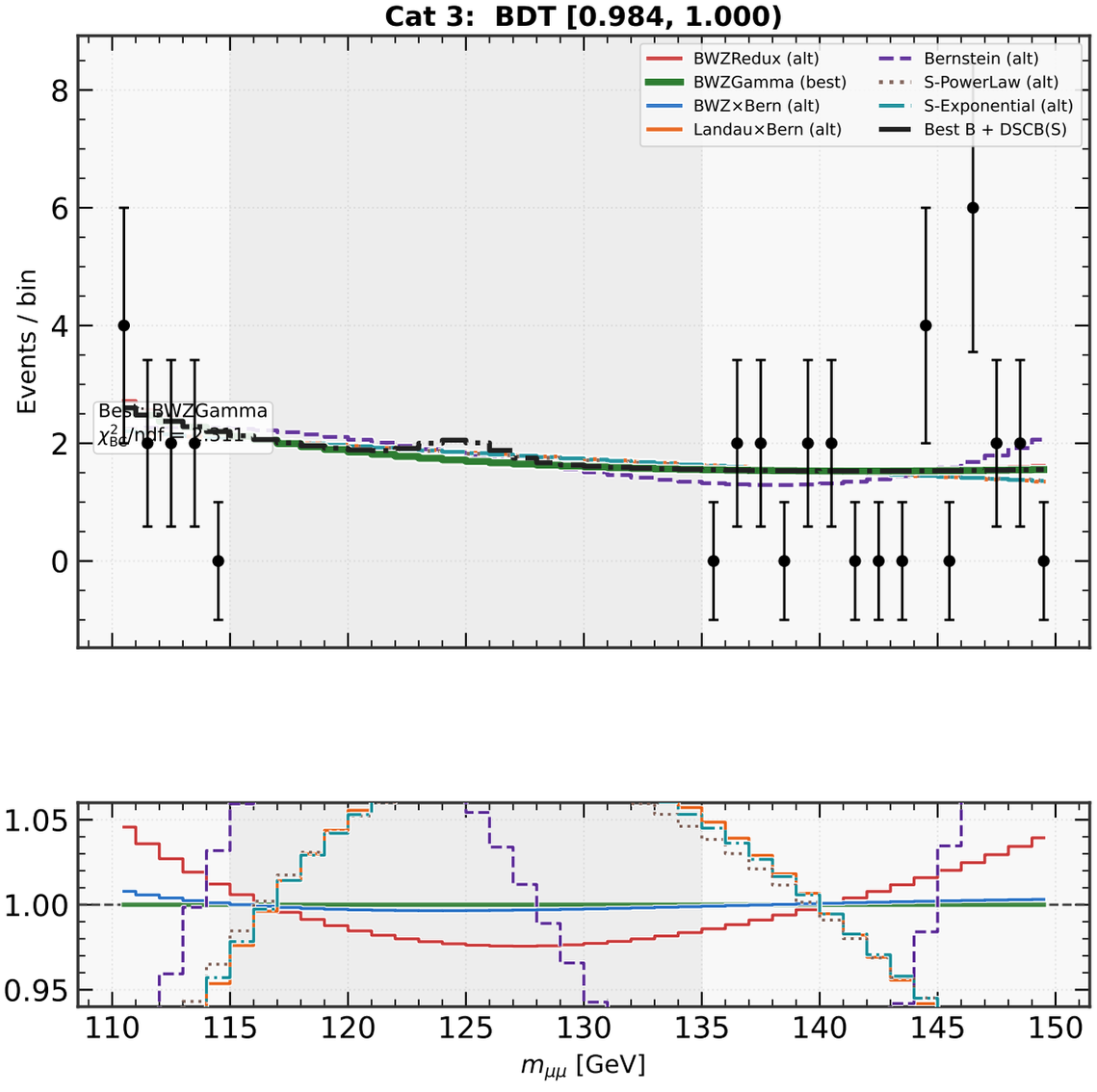
Cat 1: BDT [0.928, 0.964)



Cat 2: BDT [0.964, 0.984)



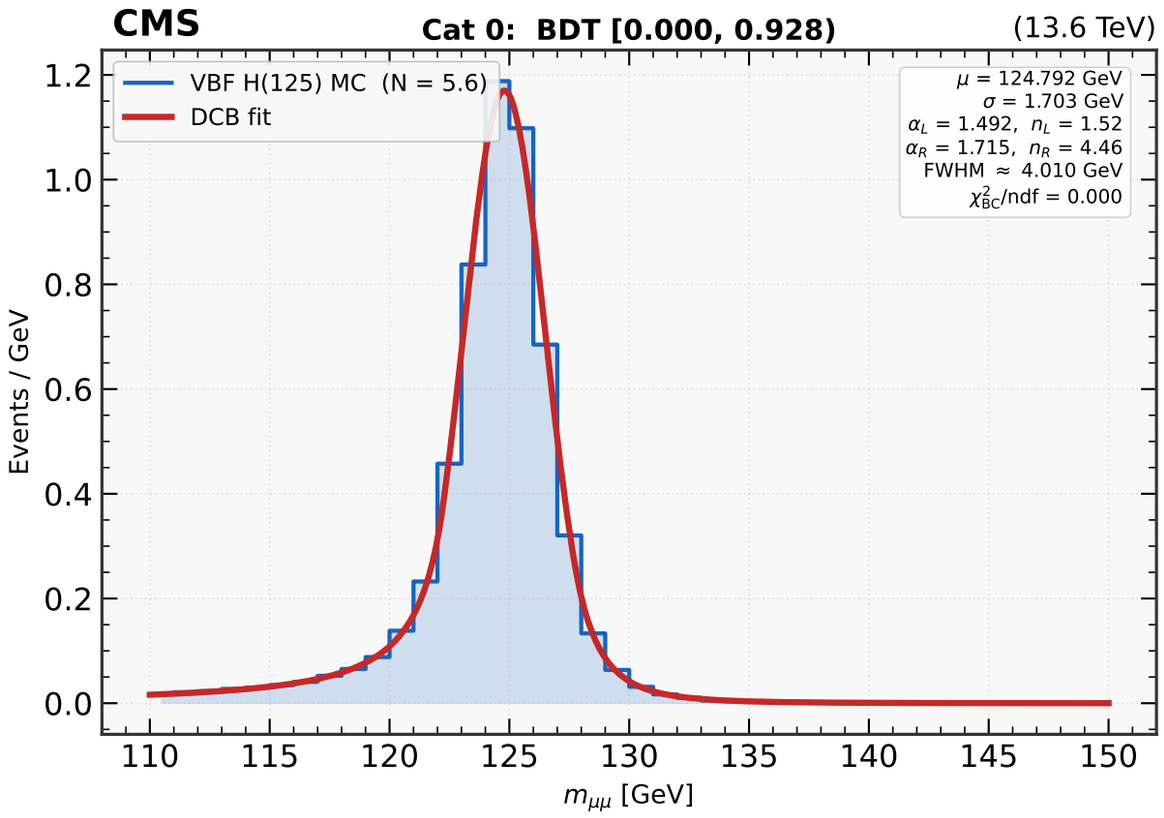
Cat 3: BDT [0.984, 1.000)

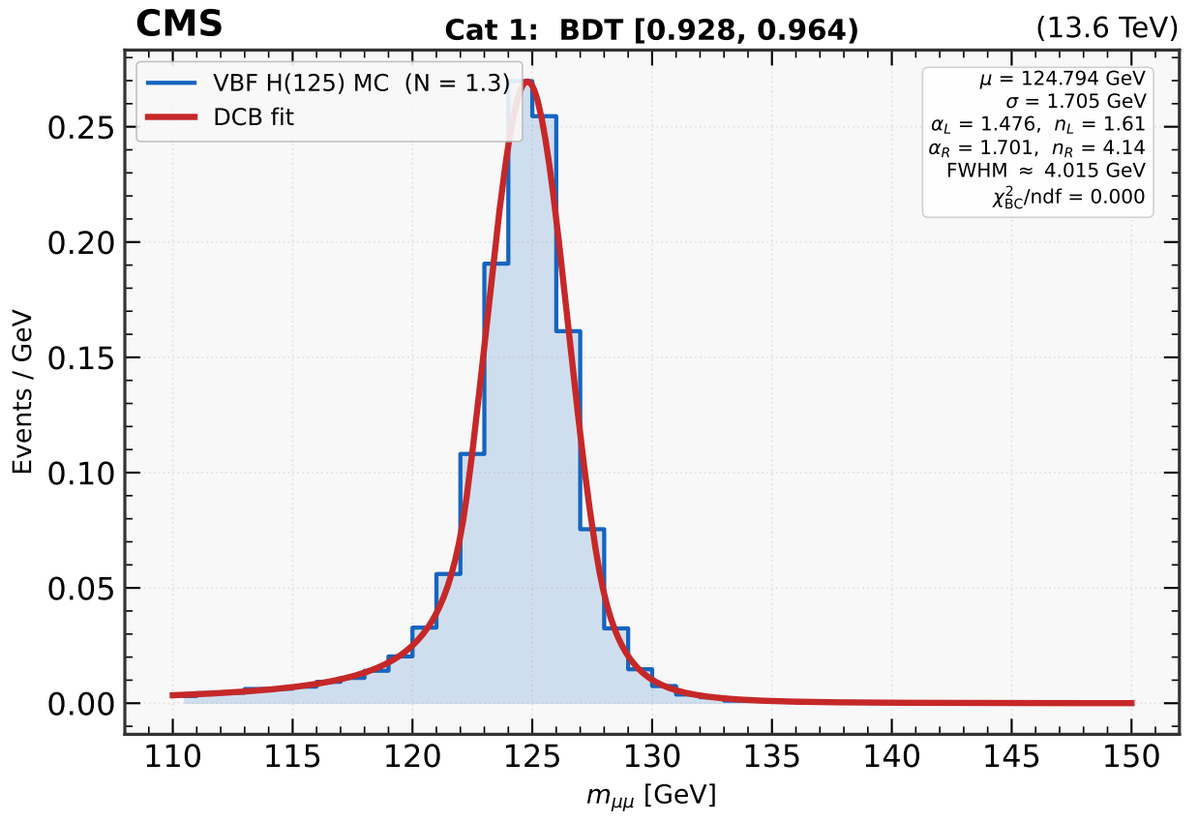


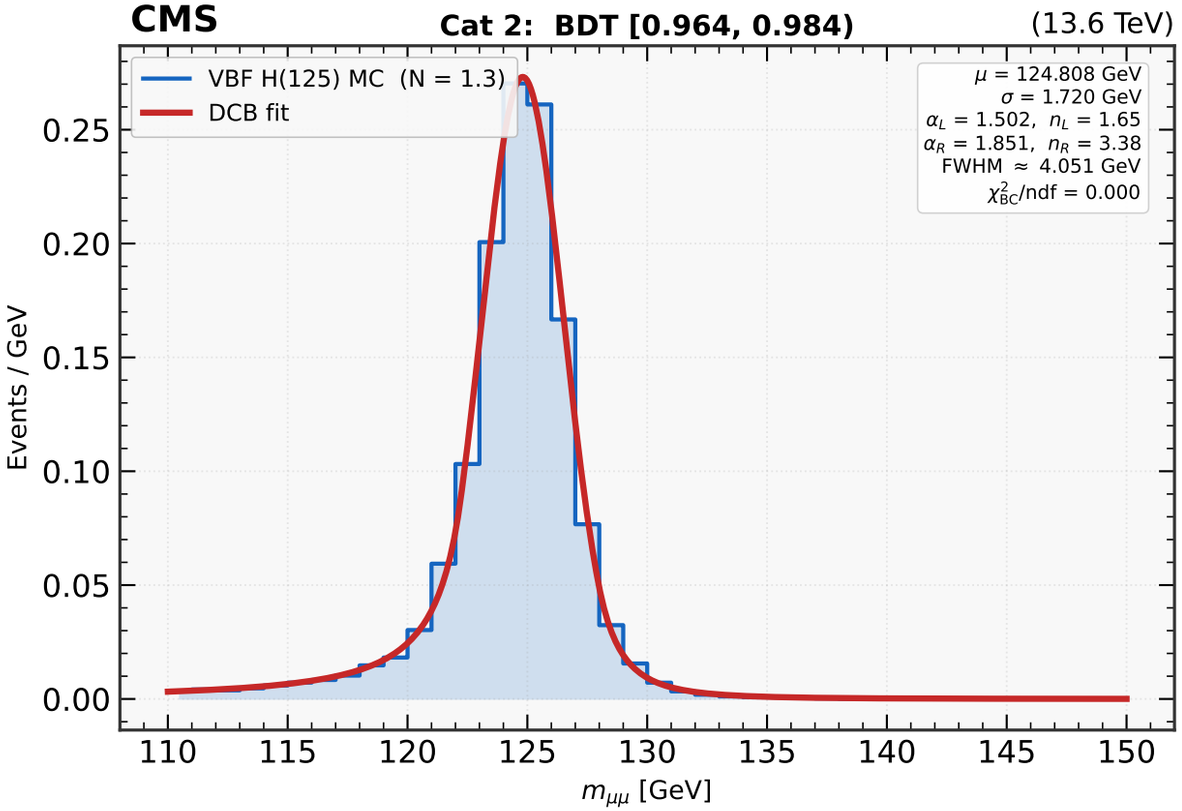
Per-category background model fit comparison (χ²_BC/ndf, ndf)

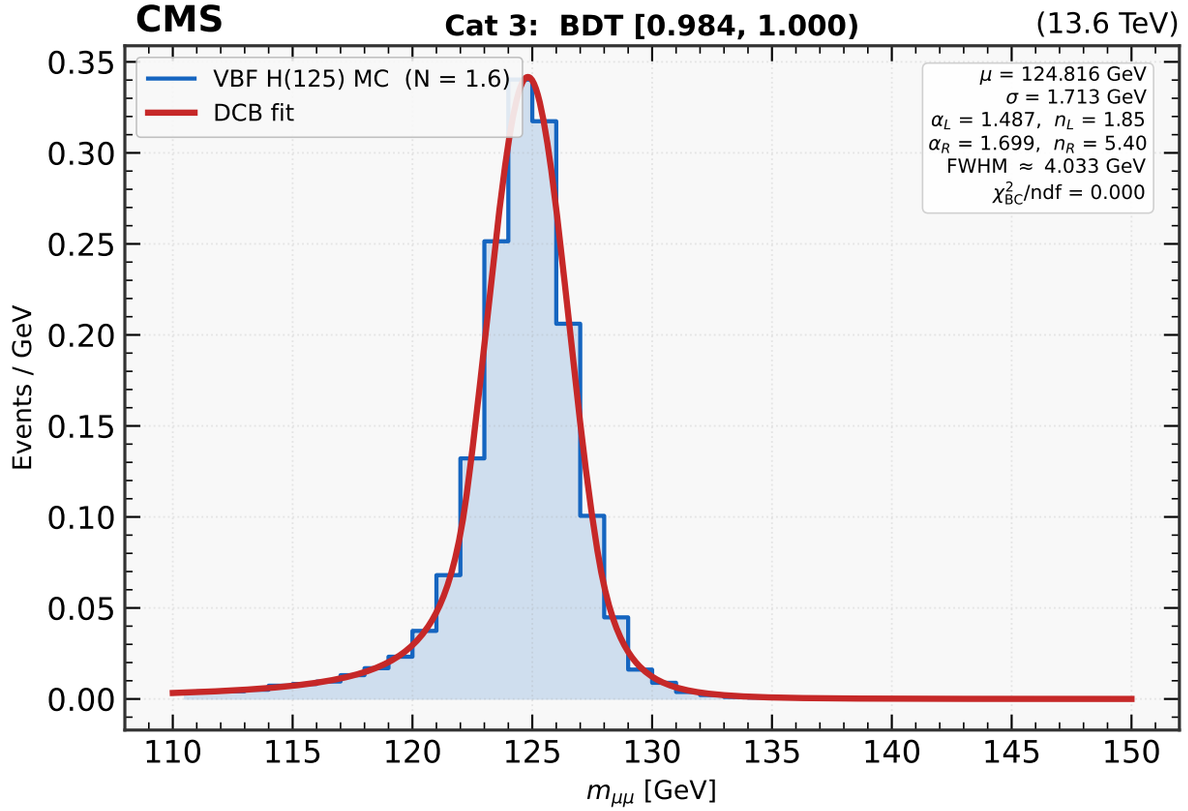
Category	BDT range	Best model	Best χ²_BC/ndf	BWZRedux	BWZGamma	BWZ×Bern	Landau×Bern	Bernstein	S-PowerLaw	S-Exponential
0	[0.000, 0.928)	BWZRedux	2.544	2.544 (37)	2.937 (37)	2.840 (36)	5.599 (37)	2.810 (35)	4.202 (37)	5.436 (37)
1	[0.928, 0.964)	BWZ×Bern	1.336	1.348 (37)	1.387 (37)	1.336 (35)	1.539 (37)	1.347 (35)	1.490 (37)	1.424 (35)
2	[0.964, 0.984)	BWZRedux	2.007	2.007 (37)	2.048 (37)	2.110 (36)	2.161 (37)	2.064 (35)	2.134 (37)	2.156 (37)
3	[0.984, 1.000)	BWZGamma	2.311	2.312 (37)	2.311 (37)	2.376 (36)	2.323 (37)	2.407 (35)	2.320 (37)	2.322 (37)

VBF H(125) signal DCB extraction per BDT category



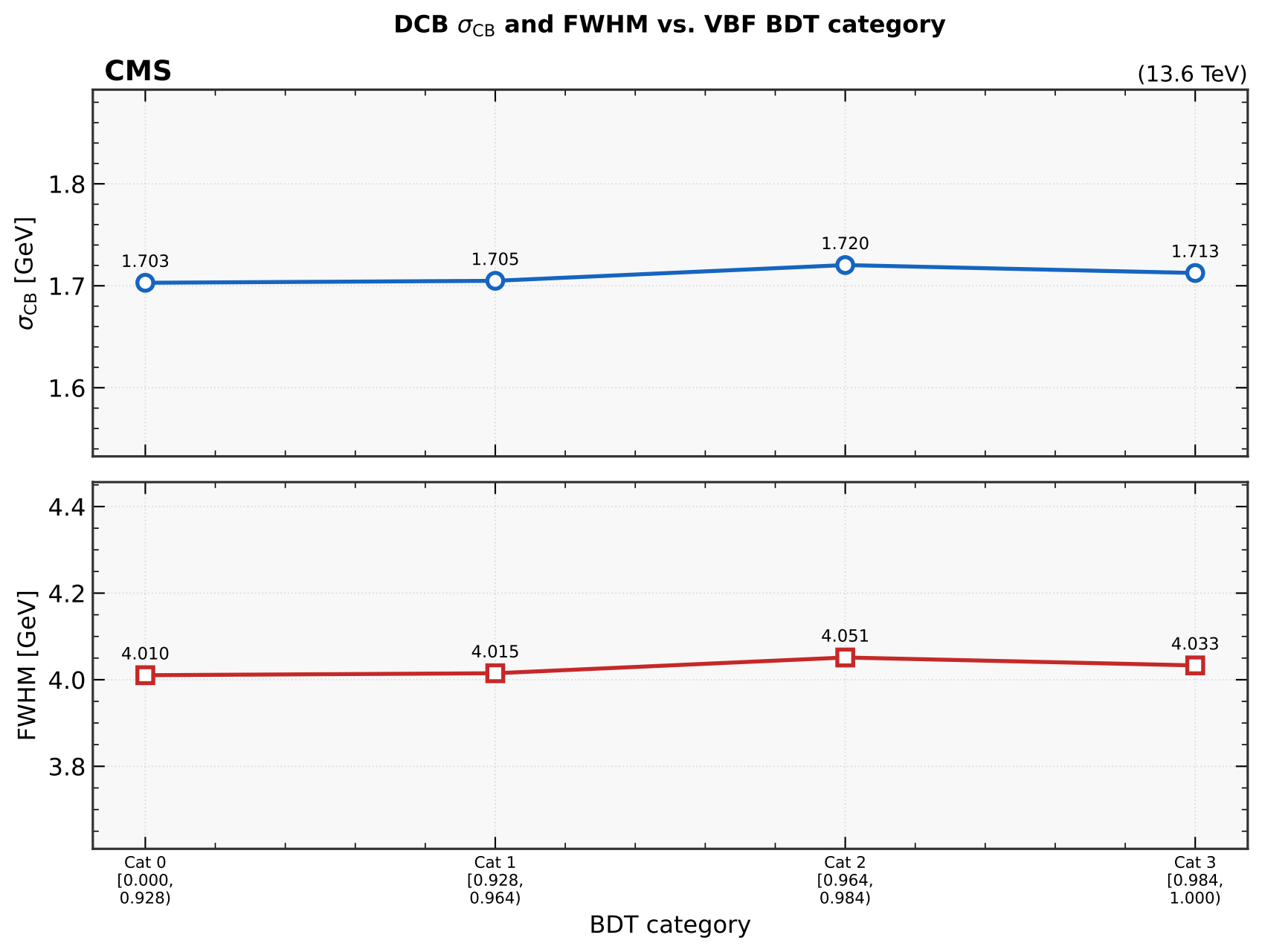


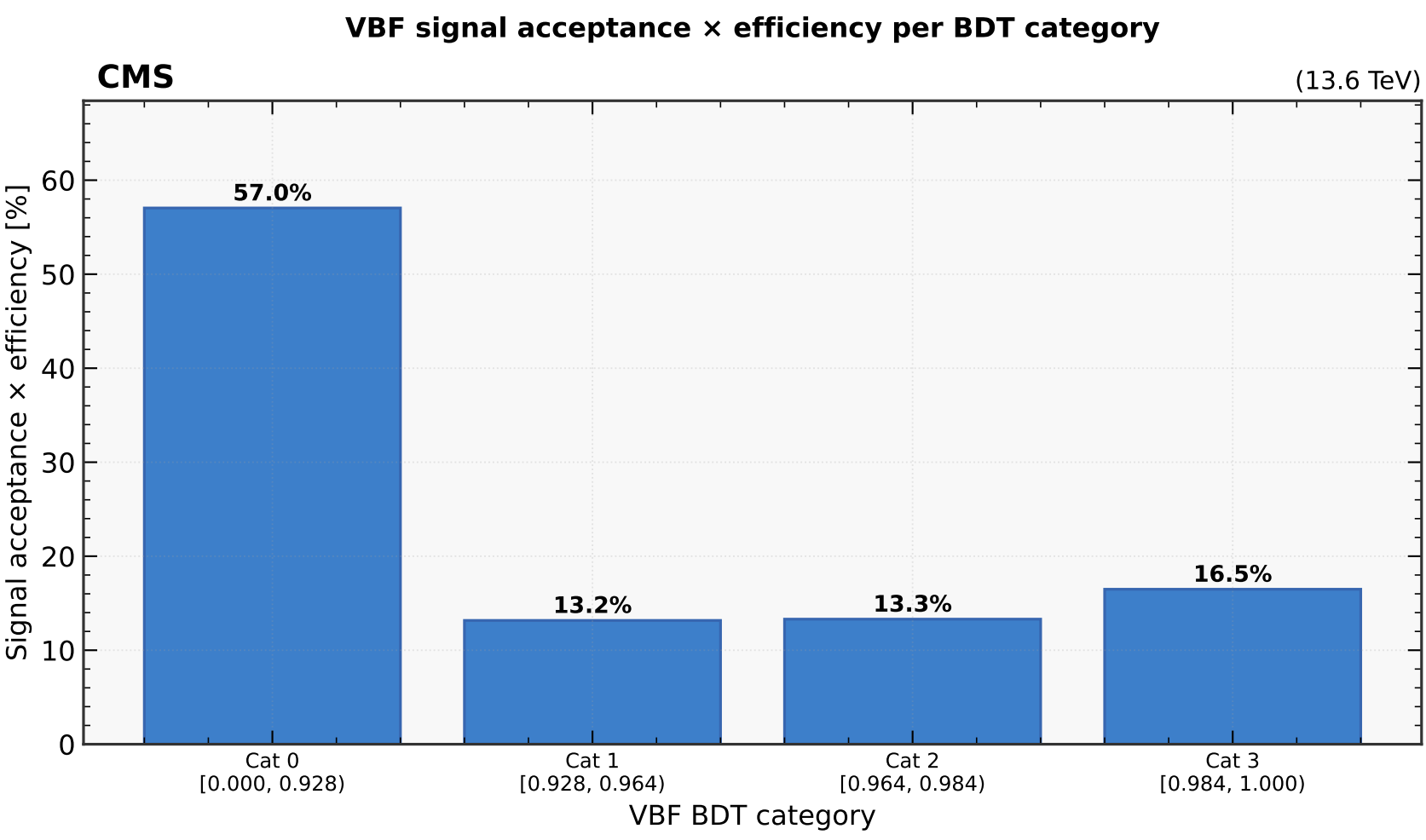




VBF signal DCB fit parameters per category

Category	BDT range	N_sig	Mean [GeV]	Sigma [GeV]	alpha_L	n_L	alpha_R	n_R	FWHM [GeV]	Asimov Z	χ^2_BC/ndf
0	[0.000, 0.928)	5.63	124.7924	1.7029	1.4924	1.517	1.7153	4.459	4.010	0.055	0.000
1	[0.928, 0.964)	1.30	124.7942	1.7049	1.4756	1.614	1.7012	4.140	4.015	0.158	0.000
2	[0.964, 0.984)	1.31	124.8084	1.7203	1.5021	1.652	1.8511	3.377	4.051	0.254	0.000
3	[0.984, 1.000)	1.63	124.8165	1.7125	1.4871	1.854	1.6993	5.404	4.033	0.703	0.000


DCB σ_{CB} and FWHM vs. VBF BDT category
CMS
(13.6 TeV)
σ_{CB} [GeV]
1.8
1.7
1.6
1.703
1.705
1.720
1.713
FWHM [GeV]
4.4
4.2
4.0
3.8
4.010
4.015
4.051
4.033
Cat 0 [0.000, 0.928)
Cat 1 [0.928, 0.964)
Cat 2 [0.964, 0.984)
Cat 3 [0.984, 1.000)
BDT category


VBF signal acceptance × efficiency per BDT category
CMS
(13.6 TeV)
Signal acceptance × efficiency [%]
0
10
20
30
40
50
60
57.0%
13.2%
13.3%
16.5%
Cat 0
[0.000, 0.928)
Cat 1
[0.928, 0.964)
Cat 2
[0.964, 0.984)
Cat 3
[0.984, 1.000)
VBF BDT category

Per-category yield and sensitivity summary

Category	BDT range	S_full	B_full	S_HWHM	B_HWHM	HWHM window [GeV]	S/√B	Asimov Z
0	[0.000, 0.928)	5.63	47770.22	3.829	4885.186	[122.8, 126.8]	0.055	0.055
1	[0.928, 0.964)	1.30	419.11	0.885	31.060	[122.8, 126.8]	0.159	0.158
2	[0.964, 0.984)	1.31	185.07	0.906	12.414	[122.8, 126.8]	0.257	0.254
3	[0.984, 1.000)	1.63	67.62	1.124	2.206	[122.8, 126.8]	0.757	0.703

Category summary: HWHM-windowed signal and background yields

Category	BDT range	Data sidebands	MC sidebands ± stat	Data/MC (sidebands)	MC full (110-150) ± stat	S/√B [HWHM]	Asimov Z (cat) [HWHM]	B_HWHM
0	[0.000, 0.928)	10068	22063.62 ± 172.12	0.456	47770.22 ± 254.12	0.0548	0.0548	4885.186
1	[0.928, 0.964)	120	190.78 ± 15.78	0.629	419.11 ± 23.72	0.1587	0.1580	31.060
2	[0.964, 0.984)	80	106.52 ± 12.24	0.751	185.07 ± 15.59	0.2571	0.2541	12.414
3	[0.984, 1.000)	32	32.71 ± 5.51	0.978	67.62 ± 8.15	0.7566	0.7031	2.206
Combined	all						0.7661	

Category composition by process (full mass window)

Category	BDT range	DY	EWK	Top	Diboson	total_err2	total
0	[0.000, 0.928)	42710.21	88.90	4901.52	69.59	64576.46	47770.22
1	[0.928, 0.964)	367.44	17.16	33.37	1.14	562.48	419.11
2	[0.964, 0.984)	155.06	15.09	14.19	0.73	242.98	185.07
3	[0.984, 1.000)	46.74	15.62	5.12	0.15	66.45	67.62